

Small language models for decentralized traffic signal control: a 2026 field guide

The 3-4B parameter class has crossed a critical threshold for real-time semantic reasoning at the edge. Models like Qwen3-4B, Phi-4-mini, and Gemma-3-4B now match or exceed the reasoning performance of 7B models from 2024 on math, logic, and instruction-following benchmarks — while fitting comfortably in under 3 GB of VRAM at Q4 quantization. This matters because a decentralized traffic signal control system placing one SLM per intersection demands models that can (a) parse structured traffic state, (b) reason about priority semantics (ambulance vs. commuter), (c) emit bounded JSON actions, and (d) do all of this within a sub-second decision budget on edge hardware. For the first time, domain-specific models with released weights exist: Traffic-R1 (3B, Apache 2.0) ([arXiv](#)) reports managing signals for **55,000+ drivers daily** in production. ([huggingface +2](#)) The critical gap is that **no model in this size class has undergone published safety-critical evaluation for real-time control**, and instruction-following reliability degrades ([Ionio](#)) **10-20% under INT4 quantization** ([Ionio](#)) ([Ionio](#)) — the very precision level needed for edge deployment.

§1. The SLM landscape shifted dramatically from 2024 to 2026

Three architectural trends converged to make SLMs plausible for control-loop reasoning. First, **Mixture-of-Experts (MoE) at small scale** — models like Granite 4.0 H-Tiny (7B total, 1B active), ([IBM](#)) NVIDIA Nemotron-3 Nano (30B total, 3.2B active), ([NVIDIA Nemotron](#)) and Gemma-4 26B-A4B route tokens through only a fraction of their parameters per forward pass, achieving large-model reasoning at small-model latency. ([Ai](#)) Second, **dual thinking/non-thinking modes** — Qwen3, Gemma 4, and Mistral Small 4 all support toggling between verbose chain-of-thought ("thinking") and fast direct-answer ("non-thinking") modes, ([arXiv](#)) ([Hugging Face](#)) letting a controller choose latency vs. accuracy per decision. Third, **distillation from frontier models** has matured: DeepSeek-R1-Distill-Qwen-7B achieves **92.8% on MATH-500** ([Hugging Face](#)) (pass@1) ([huggingface](#)) by distilling reasoning traces from the full R1 model, and LightGPT-0.5B outperforms Llama-2-70B on traffic signal control tasks ([arXiv](#)) through GPT-4 trajectory distillation. ([GitHub](#)) ([arXiv](#))

The community landscape has shifted from Llama-centric to Qwen-dominant. Sebastian Raschka's *State of LLMs 2025* report notes Qwen overtook Llama in downloads and derivatives. ([Interconnects](#)) Context windows expanded from 8-32K to 128-262K tokens as standard for SLMs. Quantization tooling matured — GGUF Q4_K_M is ([SitePoint](#)) now the de facto edge deployment format, and Google's quantization-aware training (QAT) for Gemma-3 preserves near-BF16 quality at 3× memory reduction. ([Ollama](#)) NVIDIA positioned SLMs as "the future of agentic AI" in a June 2025 position paper, ([LogRocket](#)) proposing systematic LLM-to-SLM distillation pipelines for edge deployment.

§2. Per-model deep dives

Phi-4-mini (3.8B) — Microsoft's strongest sub-4B contender

(a) Reasoning benchmarks (all vendor-reported unless noted): MMLU **67.3%**, GSM8K **88.6%**, MATH **64.0%**, BBH **70.4%**, ARC-Challenge **83.7%** (huggingface) (independently confirmed as highest among sub-10B models by awesomeagents.ai leaderboard, March 2026), (awesomeagents) HellaSwag 69.1%, GPQA 30.4%, MMLU-Pro 52.8%. (arXiv) Artificial Analysis independently measured output speed at 43.3 tok/s via Azure API with TTFT 0.82s.

(Artificial Analysis) No LMSYS Chatbot Arena Elo or Open LLM Leaderboard v2 ranking found.

(b) Instruction-following / structured output: IFEval **70.1%**, BFCL **70.3%** (arXiv) (both vendor-reported). Native function calling via `<|tool|>` tokens. (NVIDIA) JSON tool-calling format supported. Red-team testing found the model "could sometimes hallucinate function names or URLs" in function-calling scenarios. (Hugging Face) IFEval improved +19.5 points over Phi-3.5-mini, indicating substantial instruction-following gains. (arXiv)

(c) Latency at quantization: Q4_K_M file size **2.49 GB**. (Hugging Face) No published Jetson Orin benchmarks exist; proxy estimate based on similar 3.8B models suggests **~38-43 tok/s on Orin Nano Super** (INT4, MLC). Intel Xeon 6 achieved 1,955 tok/s at BF16 (1K in/1K out). (Intel) On RTX 4090, 3.8B Q4_K_M models typically achieve **~150+ tok/s decode**. N/A for TensorRT-LLM specific numbers. CPU-only inference feasible at Q4 (~3-4 GB RAM).

(d) Memory footprint: BF16 7.68 GB, Q8_0 4.08 GB, Q5_K_M 2.85 GB, **Q4_K_M 2.49 GB**.

(Hugging Face) Uses GQA with 8 KV heads. (Hugging Face) Estimated KV cache ~48 MB/1K tokens (FP16). Minimum VRAM at Q4_K_M: ~3.5 GB (2K ctx), ~4.0 GB (4K ctx), ~4.5 GB (8K ctx).

(e) License: MIT — fully permissive for commercial use, derivatives, redistribution.

(Artificial Analysis) No geographic restrictions, revenue thresholds, or output caps. Model card recommends against high-risk deployment without additional safeguards, but this is advisory, not binding. (huggingface) (NVIDIA)

(f) Safety-adjacent weaknesses: TruthfulQA (MC2) 66.4% (vendor). (arXiv) RAI defect rate 3.75% (comparable to GPT-4o-mini at 4.25%). (arXiv) Biography hallucination documented: responds to "Who is X?" with fabricated biographies. (arXiv) Multi-turn jailbreak susceptibility noted across languages. (Hugging Face) SimpleQA ~5-10% accuracy indicates minimal factual recall. (arXiv) No HaluEval/FActScore data. No published prompt injection quantitative evaluation.

(g) Safety-critical evaluations: Not evaluated in any autonomous driving, traffic control, robotics, medical, or industrial control context. Model card states models "may not be suitable for scenarios that could have consequential impact." (huggingface) (Hugging Face)

(h) Quantization degradation: No Phi-4-mini-specific Q4 vs. FP16 head-to-head. Cross-model evidence (ionio.ai study on similar architectures): Q4_K_M retains ~95% MMLU, ~84-87% GSM8K, but IFEval can drop >10%. (ionio) (ionio) Microsoft's QAT INT4 for predecessor Phi-3.5-mini improved over standard PTQ. (Hugging Face) Unsloth Dynamic Quants available for Phi-4-mini. (Hugging Face)

Phi-3.5-mini (3.8B) — predecessor baseline

Reasoning: MMLU 65.5%, GSM8K 76.9%, (Hugging Face) MATH 48.5%, BBH 63.1%, ARC-C 84.6% (all vendor-reported). IFEval only **50.6%**, (Medium) BFCL 66.1% (arXiv) — significantly weaker instruction-following than Phi-4-mini. No native function calling. MIT license. Q4 GGUF ~2.2 GB. Largely superseded by Phi-4-mini on every metric. Include only as a reference point for researchers with existing Phi-3.5 infrastructure.

Qwen2.5-7B-Instruct — the 7B reference with excellent structured output

(a) Reasoning benchmarks (vendor-reported): MMLU **74.2%**, (Qwen) GSM8K **85.4%**, MATH base 49.8% (github) / instruct **75.5%**, (Qwen) BBH **70.4%**, ARC-C 63.7%, HellaSwag 80.2%, GPQA 36.4%, MMLU-Pro 45.0%, HumanEval **84.8%**. (github +2) Significantly outperforms Gemma-2-9B-IT and Llama-3.1-8B-Instruct across all tasks except IFEval (where it slightly trails Gemma-2-9B-IT). (Qwen) (LLM Stats)

(b) Instruction-following / JSON: Qwen2.5 was **specifically designed for reliable JSON structured output** — a headline feature. (Hugging Face +2) Pre-trained with function-calling templates natively. (DeepWiki) Supports `response_format: { type: "json_object" }` via API and grammar-constrained generation in llama.cpp. (Alibaba Cloud) IFEval: slightly below Gemma-2-9B-IT (exact score not individually published). BFCL: not individually reported for 7B.

(c) Latency: Orin Nano Super (INT4, MLC): **21.75 tok/s** (NVIDIA official). (nvidia) RTX 4090 Q4_K_M: ~**104 tok/s** decode. (Ajit Singh) RTX 3060: ~42 tok/s. (Ajit Singh) TTFT scales with prompt length; at 2K prompt, expect ~200-400ms on Orin Nano. GPTQ-Int4 and AWQ variants officially available. (Read the Docs)

(d) Memory: BF16 ~15.2 GB, Q4_K_M ~**4.5-5.0 GB**. GQA with **4 KV heads** (very efficient). KV cache ~28 MB/1K tokens (FP16). Min VRAM Q4_K_M: ~4.6 GB (4K ctx), ~4.7 GB (8K ctx). Tight fit on Orin Nano 8GB (~5.2 GB usable). (ericxliu)

(e) License: Apache 2.0 — fully permissive. (github) (Hugging Face) Note: does not meet OSI Open Source AI Definition (training data/code not released), but weights are freely usable commercially. (Wikipedia)

(f) Weaknesses: TruthfulQA 56.4% (moderate). (github) Quantization-tolerant — ionio.ai independently found Qwen2.5 is "arguably the most quantization-tolerant model family" tested. (ionio) IFEval slightly weaker than Google's Gemma-2-9B-IT. (Qwen) Smaller Qwen models have degraded multilingual performance.

(g) Safety-critical: Qwen2.5-VL-7B evaluated on SafeDrive228K autonomous driving benchmark — "only slightly surpassing 50%" on SafeDrive Score. (arXiv) (arXiv) Used in VLM-AutoDrive and AgentThink research but with poor zero-shot performance requiring fine-tuning. **Not evaluated in traffic signal control.**

(h) Quantization: ionio.ai study: Q4_K_M retains ~95% MMLU, ~84-87% GSM8K, but IFEval drops >10%. AWQ outperforms GPTQ at equivalent bit-width. (ionio +2) Q5_K_M is the optimal trade-off for most tasks. (ionio)

Qwen2.5-3B-Instruct — beware the license

Benchmarks: outperforms Phi-3.5-mini and MiniCPM3-4B in math/coding (vendor claim). (Qwen) Individual benchmark scores not separately tabled. GQA with only **2 KV heads** — extremely efficient KV cache. Q4_K_M ~1.8-2.0 GB. **License: Qwen Research License — NON-COMMERCIAL ONLY.** (github +2) This is a critical restriction that disqualifies it from any production deployment. Researchers may use it freely, but any commercial traffic control system would need to switch to Qwen3-4B (Apache 2.0) instead. (Wikipedia)

Qwen3-4B — the sweet spot for edge reasoning

(a) Reasoning (vendor-reported): MMLU **76.89%** (vs. Qwen2.5-7B's 74.16%), MMLU-Pro **56.73%**, BBH **78.40%**, GPQA **44.44%**, GSM8K **89.84%**, MATH **60.80%**. (arXiv) In thinking mode: MATH-500 reaches **97.0%**. (Open Laboratory) A 4B model that **outperforms the previous-generation 7B** across the board. (arXiv) (Qwen)

(b) Instruction-following: Native tool calling supported. (Hugging Face) Thinking mode enables verbose chain-of-thought; non-thinking mode gives fast responses. (arXiv +4) Qwen3-4B-Instruct-2507 variant is non-thinking-only. (Hugging Face) (distil labs) JSON structured output inherited from Qwen2.5 design. (Hugging Face) BFCL individual score not published.

(c) Latency: Q4_K_M ~2.5 GB. No specific Orin benchmark published; proxy from architecture similarity to Phi-3.5 3B suggests **~38-43 tok/s on Orin Nano Super**. RTX 4090: ~120+ tok/s estimated. 32K native context, extensible to 131K with YaRN. (Hugging Face)

(d) Memory: BF16 ~8 GB. Q4_K_M ~2.5 GB. GQA with 4 KV heads. Min VRAM at Q4_K_M: ~3.0 GB (4K ctx). Comfortably fits Orin Nano 8GB.

(e) License: Apache 2.0 — fully permissive commercial use. (Wikipedia) (Qwen)

(f) Weaknesses: Trained via strong-to-weak distillation from larger Qwen3 models (Open Laboratory) — reasoning ceiling is inherently limited. (Open Laboratory) (arXiv) Thinking mode generates extremely verbose output (Qwen3.5-9B produced 200M tokens during Artificial Analysis evaluation vs. 23M average). (Artificial Analysis) Non-thinking mode performance drops substantially (Qwen3-32B: 95.16% → 43.55% on MATH-500 without thinking). Some llama.cpp Vulkan backend issues reported. (GitHub)

(g) Safety-critical: Not directly evaluated. Qwen3-VL-4B used as teacher model for autonomous driving label generation ([GitHub](#)) (Alpamayo-R1) and to build driving models achieving 78+ driving scores.

(h) Quantization: Qwen family is the most quantization-tolerant per independent studies. ([Ionio](#)) Q4_K_M expected ~5% MMLU loss, larger GSM8K sensitivity. Unsloth Dynamic Q4_K_XL achieves 99.9% KL divergence retention. ([Unsloth AI](#))

Qwen3-8B — upper-bound reference within the 8B ceiling

8.2B parameters, ([Siliconflow](#)) ([APXML](#)) 36 layers, 32K native context. ([Open Laboratory](#)) Outperforms Qwen2.5-14B-Base on over half of benchmarks (vendor claim). ([Open Laboratory](#)) ([arXiv](#)) MMLU ~79-80%, GSM8K ~90%+, MATH ~62%+. Apache 2.0 license. ([Open Laboratory +2](#)) Q4_K_M ~5.0 GB ([Will It Run AI](#)) — feasible on Orin Nano 8GB but tight. RTX 4090: ~104 tok/s decode. ([Ajit Singh](#)) GQA with 4 KV heads. Dual thinking/non-thinking modes. ([Siliconflow +2](#)) RULER long-context score >89 in non-thinking mode. ([Open Laboratory](#))

Qwen3.5-4B and 3.5-9B — March 2026 multimodal MoE variants

Released March 2, 2026 ([GitHub](#)) with hybrid Gated Delta Networks + sparse MoE architecture. ([GitHub](#)) Natively multimodal (text+image+video). ([Emelia](#)) **262K native context**, ([BentoML](#)) ([Unsloth AI](#)) extensible to 1M tokens. ([Hugging Face](#)) 201 languages. ([BentoML +3](#)) All Apache 2.0. Qwen3.5-4B achieves HMMT Feb 2025 score of **74.0** ([VentureBeat](#)) and "delivers performance close to the previous Qwen3-80B-A3B." ([Emelia](#)) Qwen3.5-9B claims GPQA Diamond **81.7%** and MMLU-Pro **82.5%** ([Awesome Agents +2](#)) — but XDA Developers independently warns of "a disconnect between what the benchmarks promise and what actually happens." ([XDA Developers](#)) Q4 VRAM: 3.5-4B ~3 GB, 3.5-9B ~5.1 GB. ([Compute Market](#)) ([Will It Run AI](#)) Caution: MoE/hybrid architecture may have limited inference framework support compared to dense models; verify llama.cpp compatibility before committing.

Llama-3.2-3B-Instruct — Meta's edge-optimized model

(a) Reasoning (vendor-reported): MMLU **63.4%**, GSM8K **77.7%**, MATH 48.0%, ARC-C 78.6%. ([awesomeagents](#)) ([GitHub](#))

(b) Instruction-following: IFEval **77.4%** — highest among sub-4B models at release. BFCL V2 **67.0%**. Outperforms Gemma-2-2B-IT (61.9%) and Phi-3.5-mini-IT (59.2%) on IFEval. ([Medium](#)) No native JSON mode; tool calling via custom prompt format.

(c) Latency: Orin Nano original: **27.7 tok/s**; ([Ericxliu](#)) Orin Nano Super: **43.07 tok/s** (NVIDIA official, INT4, MLC). ([nvidia](#)) RTX 4090 Q4_K_M: ~150+ tok/s. Apple M-series: ~60-100 tok/s. Very well-supported in llama.cpp. Q4_K_M file: **1.88 GB**. ([arXiv](#))

(d) Memory: BF16 ~6.0 GB, Q4_K_M **1.88 GB**. ([arXiv](#)) GQA with 8 KV heads. KV cache ~56 MB/1K tokens (FP16 estimate, 28 layers). Min VRAM Q4_K_M: ~2.5 GB (2K ctx), ~3 GB (8K ctx).

Comfortably fits Orin Nano.

(e) License: Llama 3.2 Community License — commercial use permitted, non-exclusive, worldwide, royalty-free. (GitHub) (Hugging Face) **700M+ MAU restriction** requires separate Meta license. (LushBinary) (Llama) AUP **explicitly prohibits operation of critical infrastructure and transportation technologies.** (Llama) Output distillation restricted to Llama-family models only. Not OSI-compliant.

(f) Weaknesses: MATH 48.0% indicates limited competition-level math. No built-in tools (unlike Llama-3.1-8B+). Multilingual support limited to 8 languages. (GitHub) ionio.ai found Llama models are "most vulnerable to aggressive quantization" among tested families. (Ionio)

(g) Safety-critical: License explicitly prohibits critical infrastructure and transportation use. (Llama) No published evaluations in these domains. Meta conducted CBRN uplift testing. (NVIDIA)

(h) Quantization: Meta provides official SpinQuant and QAT+LoRA versions. (Llama) SpinQuant: MMLU 63.4→62.0, GSM8K 77.7→75.7, IFEval 77.4→73.5. QAT+LoRA: best preservation (MMLU 62.4, GSM8K 77.9, IFEval 75.9). (GitHub) Q4_K_M WikiText-2 perplexity 8.57 ± 0.06 . (arXiv) Quantized models have **reduced context (8K vs. 128K).** (Llama)

Llama-3.2-1B-Instruct — the ultra-lightweight floor

MMLU 49.3% (near random on many subjects), GSM8K 44.4%, IFEval 59.5%, (GitHub) BFCL V2 **25.7%** (nearly non-functional tool calling). (Medium) Q4_K_M ~0.75 GB, runs on virtually any hardware. Same Llama Community License with critical infrastructure prohibition. SpinQuant: MMLU drops to 47.3, GSM8K to 40.6. (GitHub) Useful only as a latency/memory lower bound — reasoning capability is insufficient for safety-adjacent decision-making.

Gemma-3-4B-IT — Google's multimodal 4B entry

(a) Reasoning (vendor-reported): MMLU-Pro **43.6%**, GPQA Diamond **30.8%**, MATH **75.6%**, (ResearchGate) GSM8K **~89.2%** (Awesome Agents) (awesomeagents.ai), HumanEval **~71.3%**, (awesomeagents) (Awesome Agents) FACTS Grounding **70.1%**. (ResearchGate) Competitive with Gemma-2-27B-IT. (arXiv) HiddenMath **43.0%** (ResearchGate) (vs. Gemma-2-9B's 10.4 — massive improvement from distillation).

(b) Instruction-following: Function calling officially supported. Structured output via built-in capabilities. Multimodal: processes text and images (Hugging Face) (SigLIP encoder, 896×896 + Pan&Scan). (arxiv) (Label Rer) IFEval not individually published but described as significantly improved.

(c) Latency: Runs on Orin Nano (confirmed by NVIDIA forum users). Q4_K_M ~2.8 GB. QAT version available (~2.8 GB, 3× less than BF16). (Ollama) 128K context window. (awesomeagents +2) No specific tok/s numbers published for Orin; desktop GPU performance expected similar to other 4B models.

(d) Memory: BF16 ~8.5 GB (including 417M vision encoder). Q4_K_M ~2.8 GB. Efficient KV cache via 5:1 local:global attention ratio with 1024-token sliding window. (arxiv) (Hugging Face)

(e) License: Gemma Terms of Use — permits commercial use but Google reserves right to **remotely restrict usage** for violations. (TechCrunch) Not OSI-compliant. No MAU restriction. Does not explicitly prohibit safety-critical use (unlike Llama).

(f) Weaknesses: SimpleQA 4.0% (ResearchGate) — near-zero factual recall. MMLU-Pro 43.6% still substantially below frontier. BFloat16 required (incompatible with older GPUs — T4, RTX 20xx, V100). (Unsloth) Vision encoder always loaded even for text-only tasks (+417M params overhead).

(g) Safety-critical: Not evaluated in control-loop contexts. ShieldGemma-2 (4B safety classifier) released alongside for image safety. (Google) Gemma Terms do not explicitly prohibit transportation use.

(h) Quantization: Official QAT versions "preserve similar quality to BF16." (Ollama) Google's distillation-based training makes Gemma-3 models more quantization-tolerant than predecessors.

Gemma-3-1B-IT and Gemma-2-2B-IT — sub-3B Google entries

Gemma-3-1B: MATH **48.0%** (ResearchGate) (dramatic improvement over Gemma-2-2B's 16.0%), GSM8K ~62.8%. (awesomeagents) (Awesome Agents) Only 32K context. (arxiv) Text-only, English-only. Orin Nano: **26.33 tok/s** (Q4_K_M, Ollama). (Ericxliu) (ericxliu) BF16 ~2.0 GB, Q4_K_M ~0.8 GB. MMLU-Pro only 14.7% (awesomeagents) (ResearchGate) — near random on hard knowledge tasks. QAT version ~0.7 GB.

Gemma-2-2B: MMLU 52.2%, GSM8K 24.3%, MATH 16.0%. (Scribd) Only **8K context** — a severe limitation. HellaSwag 72.9%. (Scribd) IFEval 61.9%. No function calling. Orin Nano Super: **34.97 tok/s** (NVIDIA official). (nvidia) Now largely superseded by Gemma-3-1B and Gemma-3-4B.

DeepSeek-R1-Distill-Qwen-7B — reasoning powerhouse

(a) Reasoning: AIME 2024 **55.5%** (pass@1), MATH-500 **92.8%**, GPQA Diamond **49.1%**, LiveCodeBench **37.6%**, Codeforces rating **1189**. (Hugging Face) (huggingface) These are extraordinary reasoning scores for a 7B model — competitive with much larger models on mathematical reasoning.

(b) Instruction-following: IFEval not officially reported for distilled variants. Full R1 achieves 83.3% IF-Eval Prompt Strict. R1-0528 added function calling and JSON output support. (ComputingForGeeks) (DataCamp) **Critical concern:** distilled models exhibit catastrophic forgetting of instruction-following and tool-use capabilities when fine-tuned on reasoning-only data. (Emergent Mind) (arXiv)

(c) Latency: FP16 ~14 GB VRAM. Q4_K_M ~4.5 GB. Orin data: only AGX Orin benchmarks exist (10-17 tok/s unoptimized HF Transformers; NVIDIA claims 180.4 with vLLM). RTX 4090: ~104

tok/s at Q4_K_M (8B class). With vLLM 0.7.2: up to 4× inference speedup with FP8/INT8 on A100/H100.

(d) Memory: FP16 ~14 GB, Q4_K_M ~4.5 GB. Max generation length 32,768 tokens. (Hugging Face) (Hugging Face)

(e) License: MIT (from DeepSeek) (ComputingForGeeks) inherited over Apache 2.0 (from Qwen2.5-Math base). Fully permissive. The Llama-8B distill variant inherits Llama Community License restrictions. (Hugging Face)

(f) Weaknesses: Language mixing (Hugging Face) (Hugging Face) (English/Chinese) in outputs. Endless repetition (Hugging Face) patterns. (Hugging Face) Safety regression from reasoning-focused distillation. (Emergent Mind) Reasoning degradation up to -50 points vs. teacher on formal logic tasks for smallest variants. (Emergent Mind) **Not designed for instruction-following or structured output** — this is a reasoning specialist.

(g) Safety-critical: Not evaluated in control-loop contexts. Medical QA fine-tuned variant achieves >92% USMLE Step 1 (community work). (Emergent Mind) Safety regression makes it unsuitable without additional alignment. (Emergent Mind)

(h) Quantization: FP8 ~99% accuracy recovery. INT8 >99% for 7B. (Ionio) INT4 (W4A16) 97%+ accuracy recovery for 7B. (Red Hat) Robust to quantization for reasoning tasks, but instruction-following may degrade.

DeepSeek-R1-Distill-Qwen-1.5B — ultra-small reasoning

AIME 2024 28.9%, MATH-500 83.9%, GPQA Diamond 33.8%. (Medium) (huggingface) Remarkable reasoning for 1.5B parameters. FP16 ~3 GB, Q4 ~1 GB. MIT license. INT4 accuracy recovery drops to 94% for 1.5B (vs. 97% for 7B). (Red Hat) (arXiv) Same instruction-following concerns as the 7B variant.

Mistral-7B-v0.3 — the mature generalist

MMLU 63.5%, HellaSwag 83.0%, GSM8K 34.5- (LLM Explorer) 47.5% (weak on multi-step math), MATH ~3% (Level 5 — essentially non-functional). TruthfulQA 42.15%. (LLM Explorer) JSON mode supported via (response_format) parameter in Instruct v0.2+. 32K context. Apache 2.0 license. FP16 ~14.5 GB, (LLM Explorer) Q4_K_M ~4.5 GB. RTX 3060: ~40-60 tok/s. Hallucination rate ~15% on factual queries. Invents URLs, dates, citations. (Ucstrategies News) No safety guardrails in base model. Now significantly outperformed by Qwen3-8B and DeepSeek-R1-Distill-7B on reasoning tasks. Retain only as a baseline reference for researchers already invested in the Mistral ecosystem.

IBM Granite 4.0 H-Tiny — enterprise MoE with exceptional instruction-following

7B total parameters, (Hugging Face) 1B active via hybrid Mamba-2/Transformer MoE. (LLM Stats)

(IBM) IFEval **84.32%** (AI Models) — the highest instruction-following score among models surveyed. GSM8K **81.35%**, MMLU **67.43%**, BBH **69.36%**, HumanEval **81**, SALAD-Bench **96.28%** (safety). (AI Models) **Apache 2.0** license, ISO 42001 certified, cryptographically signed. (IBM) BFCLv3: keeps pace with much larger models. (IBM) Over **70% lower memory** and 2× faster inference vs. similar-sized conventional transformers (IBM) due to Mamba-2's linear-time sequence processing. Granite Guardian companion models available for hallucination/jailbreak detection. (IBM) Caveat: hybrid Mamba-2 architecture has **limited inference framework support** — verify llama.cpp/TensorRT-LLM compatibility before adopting.

SmolLM2-1.7B — HuggingFace's ultra-compact model

Trained on 11T tokens (FineWeb-Edu, DCLM, The Stack). (Hugging Face) BFCL **27%** (weak tool calling). (Hugging Face) Apache 2.0. BF16 3.4 GB, Q4_K_M ~**1.2 GB**. 8K context (16K with extended version). English-focused. Ranked #9 on awesomeagents.ai SLM Leaderboard. SFT + DPO alignment. Supports function calling with structured format. Reasoning capability significantly below the 3-4B class but viable for extremely memory-constrained scenarios.

Falcon3-7B and Falcon-H1 series — emerging competitors

Falcon3-7B: ARC-C 65.9%, GSM8K 79.1%. (Falcon) Trained on 14T tokens. (Hugging Face) 4 languages (EN, FR, ES, PT). 32K context. (Hugging Face) Apache 2.0 for newer variants. GGUF, GPTQ-Int4, AWQ available. (Middleeastainews) **Falcon-H1R-7B** (hybrid Mamba/Transformer, 2025) achieves AIME 2025 **89.3%** and GPQA-D 61.3% with FP8 (Falcon) — remarkable reasoning, but limited ecosystem support compared to Qwen/Llama.

Traffic-R1 (3B) — the domain-specific specialist

(a) Reasoning: State-of-the-art zero-shot on CityFlow TSC benchmarks (ATT, AWT, AQL). Outperforms MaxPressure, MPLight, CoLight, AdvancedCoLight, and LLMLight. (arXiv) (ResearchGate) Handles out-of-distribution incidents (sensor failures, road closures, emergency vehicles) where RL baselines completely fail. (huggingface +2)

(b) Instruction-following: Compatible with LLMLight evaluation code with prompt format modifications. (huggingface) (Hugging Face) Uses structured text-based input/output for phase selection. Not benchmarked on general IFEval/BFCL.

(c) Latency: 3B parameters (arXiv) on Qwen2.5-3B base. "Lightweight enough for real-time inference on mobile-class chips." (huggingface) (arXiv) Q4_K_M ~1.8-2.0 GB estimated. No published tok/s numbers.

(d) Memory: ~6 GB FP16. Edge-deployable at Q4. One quantized version on HuggingFace.

(e) License: Apache 2.0.

(f) Critical caveat: The released checkpoint is "**Public 0.1**" — an **early version** that intentionally excludes some offline pretraining data due to "commercial and privacy concerns." (huggingface)

Hugging Face The full production model is internal. Prompt format is inconsistent with LLMLight's default — requires code modification. huggingface Hugging Face No independent third-party validation. Team is upgrading base to Qwen3/VL. Hugging Face

- (g) **Safety-critical:** Evaluated for OOD incident handling but no formal adversarial robustness, hallucination rate, or safety-specific benchmarks reported. **Production deployment:** managing signals for 55,000+ drivers daily, reducing average queue lengths >5%.
- (h) **Quantization:** One quantized model exists; no degradation metrics published.

LightGPT family — distilled TSC specialists

Available in multiple sizes: Qwen2-0.5B, Llama2-7B, Llama2-13B, Llama3-8B variants. MIT license. Published at KDD 2025 (Audience Appreciation Award). Key result: **LightGPT-0.5B outperforms Llama2-70B** on TSC tasks — a 140× parameter efficiency gain through domain-specific distillation from GPT-4 reasoning trajectories + RL critic refinement. On 4×A800-80GB: LightGPT-8B handles 20 intersections in parallel, up to 100 asynchronously. Code fully open-sourced at github.com/usail-hkust/LLMTSCS.

§3. Domain-adapted methods without released weights

The following systems use LLMs for traffic signal control but do not release deployable model weights. They serve as **benchmark anchors** — comparison points for what reasoning-augmented TSC can achieve, not candidates for your deployment.

System	ArXiv ID	Base Model	Released Weights	Status
LLMLight	2312.16044	GPT-4/3.5 + LightGPT	LightGPT only	KDD 2025
CoLLMLight	2503.11739	8B LLM + GPT-4o	✗	Under review
LA-Light	2403.08337	GPT-4 + LangChain	✗	Published 2024
TrafficGPT	2309.06719	gpt-35-turbo	✗	Transport Policy 2024
VLMLight	2505.19486	Qwen2.5-VL-32B + 72B	✗	Under review
iLLM-TSC	2407.06025	GPT-4	✗	Published 2024
CuraLight	2604.05663	Gemma-3 + GPT	✗	Published Apr 2026
SignalClaw	2604.05535	LLMs (evolution)	✗	Published Apr 2026

LLMLight pioneered single-intersection LLM-based TSC. Each intersection receives a prompt containing real-time queue lengths, vehicle counts, scenario description, commonsense knowledge (e.g., "prioritize long queues"), and action space. GPT-4 with commonsense prompting achieves state-of-the-art or competitive results across Hangzhou 4×4, Jinan 3×4, New York, and synthetic CityFlow scenarios. The framework is fully reproducible with open code; GPT-4 API costs are the main barrier.

CoLLMLight extends LLMLight to cooperative multi-intersection control — the first LLM framework for network-wide TSC. It builds structured spatiotemporal graphs capturing inter-intersection relationships and uses three-level complexity-aware reasoning. An asynchronous architecture decouples slow LLM reasoning from fast signal switching. CoLLMLight-8B outperforms general-purpose 32B and 70B models on TSC tasks, and consistently beats LLMLight on complex networks like NYC 28×7. Code is open-sourced; model weights are not released.

VLMLight introduces vision-language reasoning for TSC. A dual-branch architecture handles routine traffic via fast RL policy and safety-critical scenarios (emergency vehicles) via structured multi-agent LLM deliberation using Qwen2.5-VL-32B + Qwen2.5-72B. Emergency vehicle waiting time reduced by **up to 65%** vs. RL-only baselines, with <1% ATT degradation on routine traffic. Inference latency for the deliberative branch fits within a standard 13-second signal phase buffer. The system uses open-weight Qwen models but requires substantial GPU resources and does not release fine-tuned checkpoints.

CuraLight (April 2026) combines diffusion-based RL exploration with multi-LLM ensemble deliberation and adversarial debate for candidate timing evaluation. Achieves **~28.5% average improvement** over base Gemma-3 across Jinan, Hangzhou, and Yizhuang networks.

SignalClaw (April 2026) takes a fundamentally different approach: LLMs serve as evolutionary mutation operators to synthesize compact, interpretable control programs (12–20 AST nodes) that match or exceed RL baselines while being fully human-inspectable.

§4. Decision matrix across three deployment scenarios

Scenario A: Maximum reasoning quality regardless of cost or latency

This scenario suits researchers who want to understand the upper bound of SLM reasoning for TSC, running evaluations on server GPUs or cloud instances where latency and memory are not binding constraints.

Model	MATH	GSM8K	MMLU	GPQA	IFEval	Structured Output	Safety-Critical Eval	Overall Reasoning
DeepSeek-R1-Distill-Qwen-7B	92.8	N/R	N/R	49.1	N/R	Weak (forgetting)	None	★★★★★
Qwen3-8B (thinking)	~62+	~90+	~80	N/R	N/R	Strong (native)	None	★★★★☆
Qwen3-4B (thinking)	97.0+	89.8	76.9	44.4	N/R	Strong (native)	Indirect (AD labels)	★★★★☆
Gemma-3-4B-IT	75.6	~89.2	43.6‡	30.8	N/R	Good (native FC)	None	★★★☆☆
Phi-4-mini	64.0	88.6	67.3	30.4	70.1	Good (native FC)	None	★★★☆☆
IBM Granite 4.0 H-Tiny	N/R	81.4	67.4	N/R	84.3	Excellent	Safety-focused org	★★★☆☆
Llama-3.2-3B	48.0	77.7	63.4	N/R	77.4	Adequate	None (prohibited)	★★☆☆☆
Traffic-R1 (3B)	N/R	N/R	N/R	N/R	N/R	TSC-specific	Production (55K/day)	Domain ★★★★★

‡MATH-500 in thinking mode. #MMLU-Pro, not MMLU.

Key trade-off: DeepSeek-R1-Distill-Qwen-7B has the strongest raw reasoning but weak instruction-following — it may not reliably produce bounded JSON output. Qwen3-4B offers the best balance of reasoning + structured output. Traffic-R1 dominates on domain-specific TSC tasks but has no general reasoning benchmarks. **IBM Granite 4.0 H-Tiny achieves the highest IFEval score (84.3%) — critical for the JSON action-output reliability this use case demands** — but its hybrid Mamba architecture requires framework verification.

Scenario B: Best balance within a 500ms end-to-end decision budget

The budget breakdown: prompt assembly (~10ms) + TTFT (dominant) + decode for ~50-token JSON output + parsing/validation (~5ms). On different hardware:

Model	Hardware	TTFT (est.)	Decode 50 tok	Total (est.)	Fits 500ms?	Reasoning
Llama-3.2-3B Q4	RTX 4090	~20ms	~330ms	~365ms	✓	Medium
Phi-4-mini Q4	RTX 4090	~25ms	~330ms	~370ms	✓	Medium- High
Qwen3-4B Q4	RTX 4090	~25ms	~400ms	~440ms	✓	High
Qwen2.5-7B Q4	RTX 4090	~40ms	~480ms	~535ms	⚠ Marginal	High
Llama-3.2-3B Q4	Orin Nano Super	~150ms	~1160ms	~1325ms	✗	Medium
Phi-4-mini Q4	Orin Nano Super	~200ms	~1310ms	~1525ms	✗	Medium- High
SmolLM2-1.7B Q4	Orin Nano Super	~80ms	~770ms	~865ms	✗	Low
Gemma-3-1B QAT	Orin Nano Super	~70ms	~1900ms*	~1985ms	✗	Low

*At 26 tok/s.

The 500ms budget is achievable on desktop/server GPUs for 3-4B models but is not achievable on Orin Nano for any model generating 50 tokens. For SUMO simulation (where the researcher controls the hardware), an RTX 4090 easily serves the budget. For speculative edge deployment, two mitigations exist: (1) **prefix caching** can reduce TTFT by 60-80% for repeated prompt templates, and (2) **speculative decoding** can improve effective throughput by 1.5-2×. Both add implementation complexity. Another option: reduce output to a single-token phase ID rather than full JSON, slashing decode cost to near-zero.

Pareto frontier (reasoning quality / latency): Qwen3-4B on RTX 4090 sits on the frontier — highest reasoning per millisecond spent. Phi-4-mini is a close second with better-documented IFEval. For Orin Nano deployment, only a single-token output design with sub-2B models approaches feasibility.

Scenario C: Most parameter-efficient for simulation-grounded research

This scenario prioritizes small footprint, easy reproducibility, permissive license, and sufficient

reasoning for meaningful SUMO experiments.

Model	Params	Q4 Size	License	MMLU	GSM8K	JSON Capable	Reproducibility	Score
Qwen3-4B	4B	~2.5 GB	Apache 2.0	76.9	89.8	✔ Native	Excellent	★★★★★
Phi-4-mini	3.8B	2.49 GB	MIT	67.3	88.6	✔ Native	Excellent	★★★★☆
Llama-3.2- 3B	3B	1.88 GB	Llama CL†	63.4	77.7	⚠ Prompt- only	Excellent	★★★☆☆
Gemma-3- 4B	~4.3B	~2.8 GB	Gemma ToU	43.6 [‡]	89.2	✔ Native	Good	★★★☆☆
Traffic-R1	3B	~2 GB	Apache 2.0	N/R	N/R	TSC- specific	Limited§	★★★☆☆
DeepSeek- R1-Distill- 1.5B	1.5B	~1 GB	MIT	N/R	N/R	⚠ Weak	Good	★★☆☆☆
SmolLM2- 1.7B	1.7B	~1.2 GB	Apache 2.0	N/R	N/R	⚠ Weak	Excellent	★★☆☆☆

[†]MMLU-Pro. [‡]Prohibits critical infrastructure use. [§]Only "Public 0.1" checkpoint released.

Qwen3-4B emerges as the strongest candidate for simulation research: it offers the best reasoning-per-parameter ratio among models with permissive licenses, native structured output, and dual thinking/non-thinking modes (useful for ablation studies). Phi-4-mini is the close alternative with stronger IFEval documentation and MIT license simplicity. Traffic-R1's domain-specific capability is compelling but the "Public 0.1" limitation and lack of general benchmarks make it harder to integrate into a rigorous experimental framework. **Llama-3.2-3B's license explicitly prohibits transportation infrastructure operation** — acceptable for pure SUMO simulation research but a non-starter for any path toward deployment.

§5. Critical gaps, caveats, and what researchers must do themselves

No SLM has been formally evaluated in safety-critical control loops

This is the single most important gap. Traffic-R1 reports production deployment but without published adversarial robustness testing, formal verification, or safety-case documentation. Every other model in this survey has **zero published evaluation** in autonomous driving, traffic control, robotics control loops, or industrial control contexts where bounded-latency decisions affect physical safety. Researchers should treat all SLM-based TSC as experimental and pair any SLM controller with a deterministic fallback (e.g., MaxPressure) that activates when model output is malformed, late, or low-confidence.

Benchmarks that mislead

Several commonly cited benchmarks have known problems. **TruthfulQA is saturated** — HalluLens (ACL 2025) found it is now included in training data, contains incorrect gold answers, and its metrics excessively penalize models. **GPQA Diamond has only 198 questions**, making vendor claims of high scores statistically noisy. The **disconnect between benchmark performance and real-world capability** is especially pronounced for SLMs: XDA Developers independently warned that Qwen3.5-9B's benchmark-topping numbers do not translate reliably to practical tasks. Researchers should not select models based solely on aggregate benchmark scores.

What you must test yourself

Three evaluations are non-negotiable for this use case and cannot be answered by existing benchmarks:

- **JSON output reliability under your exact prompt template.** IFEval measures general instruction-following, not your specific `{phase: "NSGT", priority: "emergency", confidence: 0.92}` schema. Run 1,000+ diverse traffic states through each candidate model at Q4_K_M and measure: valid JSON rate, schema compliance rate, semantic correctness rate, and failure mode distribution (malformed JSON vs. valid-but-wrong vs. refusal).
- **Latency profiling on your target hardware at your context length.** Published benchmarks use different prompt lengths, batch sizes, and frameworks. Profile TTFT and decode tok/s with your actual prompt template (likely 500–2000 tokens input, 30–80 tokens output) on your actual GPU, using your actual inference stack.
- **Graceful degradation under adversarial traffic states.** Construct edge cases: simultaneous emergency vehicles, sensor dropout, contradictory data, prompt injection via crafted vehicle metadata. No published benchmark covers this. Measure whether the model outputs safe fallback actions or hallucinates dangerous phase selections.

The Traffic-R1 question

Traffic-R1 demonstrates that domain-specific RL fine-tuning on a 3B model can outperform both general-purpose SLMs and traditional RL controllers on TSC benchmarks, including out-of-distribution scenarios. LightGPT's 0.5B variant outperforming Llama-2-70B provides further evidence that **domain-specific distillation yields 100×+ parameter efficiency gains** for narrow tasks. However, the released "Public 0.1" checkpoint is explicitly incomplete, the prompt format requires code modification to work with LLMLight's evaluation framework, and no independent replication exists. The team is actively upgrading to Qwen3/VL base models, which may render the current checkpoint obsolete. For dissertation research, the safest strategy is to benchmark Traffic-R1's public checkpoint alongside a well-prompted Qwen3-4B (using the LLMLight prompting framework) and a fine-tuned LightGPT variant, measuring whether domain specialization provides statistically significant gains on your specific scenarios.

Quantization degrades what matters most for control loops

The cross-cutting finding from quantization research is sobering: **Q4_K_M preserves ~95% of MMLU but can degrade IFEval by >10% and GSM8K by >13%**. For a traffic signal control loop, instruction-following reliability (will the model emit valid JSON?) and numerical reasoning (will it correctly compare queue lengths?) are more important than general knowledge. These are precisely the capabilities most damaged by aggressive quantization. The ionio.ai study found AWQ consistently outperforms GPTQ at equivalent bit-width. Google's QAT approach for Gemma-3 and Meta's QAT+LoRA for Llama-3.2 both show superior preservation over post-training quantization. **Q5_K_M is the recommended minimum precision** for reasoning-critical applications; drop to Q4_K_M only if memory constraints are truly binding.

Prompt injection is an architectural vulnerability, not a model-specific one

OWASP ranks prompt injection as the #1 LLM vulnerability for 2025. Attack success rates of **76–90%** are reported across model families. For a decentralized traffic control system, an adversary who can inject data into the traffic state feed (e.g., spoofed vehicle IDs claiming emergency status) could manipulate phase selections across the network. No model in this survey has demonstrated robust prompt injection resistance at the small scale. **This is a system-design problem, not a model-selection problem** — it requires input validation, output constraint enforcement (grammar-constrained decoding), and deterministic safety bounds independent of the LLM.

Models not yet released or unverifiable as of April 2026

Qwen3.5-4B and 3.5-9B (March 2026) use a hybrid Gated Delta Networks + sparse MoE architecture with limited llama.cpp support verification. IBM Granite 4.0 H-Tiny's hybrid Mamba-2 architecture similarly lacks confirmed broad inference framework compatibility. The NVIDIA Nemotron-3 Nano (30B/3.2B active) is architecturally compelling but its 30B total parameters place it outside typical SLM memory budgets. Researchers should verify inference

framework support for any non-standard architecture before committing to a model for their experimental pipeline.